Non-clinical Study Summary

Test article TRN-416 — in vitro pharmacology, binding and selectivity

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Document owner	Non-clinical Sciences
Test article	TRN-416 (anti-human TNFRSF1B monoclonal antibody)
Studies covered	TRN416-FC-002 to FC-008, FC-012A, FC-012B, EL-012C, AR-012D, MB-012E, RS-001, BP-009, BP-010, BP-011
Donor consent protocol	HS-2019-114
Animal use protocol	IACUC-2016-041
Scope	Methods and conditions as performed. Tabulated results are held in the accompanying data files.

1. Test article identity

Designation	TRN-416
Target	Human tumour necrosis factor receptor superfamily member 1B (TNFRSF1B, TNFR2)
Species of origin	Mouse. Hybridoma raised in BALB/c.
Isotype	IgG1, kappa
Immunogen	Recombinant fragment of the human TNFR2 extracellular region spanning amino acids 50 to 122, followed by boosting with the whole exterior portion of the human TNFR2 protein.
Immunisation schedule	Intravenous and subcutaneous administration on at least five occasions before splenic harvest. Tail bleed titres were checked by ELISA; animals reaching a titre between 1:30,000 and 1:100,000 were carried forward to hybridoma generation.
Hybridoma history	Generated at a contract hybridoma facility. Supernatants were screened first; subclones retaining activity were purified. Clones of the IgM isotype were discarded.
Purification	Protein A affinity chromatography followed by buffer exchange into phosphate-buffered saline.
Purity	Greater than 95% by sodium dodecyl sulfate polyacrylamide gel electrophoresis and size-exclusion chromatography.
Endotoxin	Determined with the Endosafe-PTS system and Endosafe PTS cartridges. Levels were below 1 EU/mg for every lot released to in vitro testing.
Concentration determination	Absorbance at 280 nm, calculated with an extinction coefficient of 205,500 M^{-1} cm^{-1} (1.0 mg/ml = A280 1.40).
Storage	Phosphate-buffered saline, pH 7.4, at 4 C for working stocks and -80 C for archive aliquots. Working stocks were not subjected to more than three freeze-thaw cycles.
Lots used	TRN416-L07, TRN416-L09, TRN416-L11

2. Intervention agents

The following agents were added to cultures in one or more of the studies described below. Each entry gives the identity of the agent and the molecular interaction through which it acts.

Recombinant human IL-2	Interleukin-2, purchased from a commercial supplier. Binds the interleukin-2 receptor. Signalling proceeds through the IL-2 receptor beta chain (CD122) and the common gamma chain (CD132), activating JAK1 and JAK3 and phosphorylating STAT5.
Recombinant human TNF	Tumour necrosis factor, purchased from a commercial supplier. A soluble trimer that binds both TNFR1 (TNFRSF1A) and TNFR2 (TNFRSF1B). TNFR1 carries an intracellular death domain and recruits TRADD; TNFR2 lacks a death domain and recruits TRAF2-containing complexes.

Anti-CD3 monoclonal antibody, clone HIT3a	Mouse IgG2a against the epsilon chain of the human CD3 complex. Binding cross-links the CD3 complex associated with the T cell receptor and initiates proximal T cell receptor signalling through LCK and ZAP-70.
Anti-CD3 monoclonal antibody, clone OKT3	Mouse IgG2a against the epsilon chain of the human CD3 complex, used plate-bound. Acts through the same proximal T cell receptor signalling route as clone HIT3a.
Cross-linking anti-rodent IgG, clone ab9165	Polyclonal antibody raised against rodent immunoglobulin G. Binds the Fc portion of murine IgG and multimerises murine antibody that is already bound to its antigen.
TRN-416	See section 1. Binds the human TNFR2 extracellular region at the sites given in section 12.

3. Donor material and cell isolation

Peripheral whole blood was drawn into K2-EDTA (purple top) tubes. Donors provided written informed consent under protocol HS-2019-114. Donor material is described in each study section as nondiabetic blood donors or as donors with type 1 diabetes.

3.1 CD4+ T cell isolation

Whole blood was washed twice with 1x Hanks' balanced salt solution containing 2% fetal bovine serum. CD4+ T cells were isolated by immunomagnetic negative selection using the EasySep Direct Human CD4+ T Cell Isolation Kit, following the instructions of the manufacturer. In brief, 1200 microlitres of Isolation Cocktail and 1200 microlitres of RapidSpheres were combined with 24 ml of whole blood in a 50 ml centrifuge tube and incubated for 5 minutes at room temperature. Twenty-four millilitres of calcium- and magnesium-free phosphate-buffered saline were added and the tube was placed in an Easy 50 EasySep magnet, immobilising the unwanted cells against the wall of the tube. After 10 minutes the enriched suspension was transferred to a fresh tube and the magnetic separation was repeated for 5 minutes with fresh RapidSpheres. The suspension was transferred again and purified a third time in the magnet.

The purity of the final CD4+ T cell preparation was greater than 95% as assessed by flow cytometry. Cell recovery was counted on a hemocytometer with trypan blue exclusion; viability was above 92% in all preparations released to study.

3.2 Peripheral blood mononuclear cells

Where peripheral blood mononuclear cells were required, they were collected on the day of venipuncture by density gradient separation over Ficoll-Hypaque Plus, centrifuged at 400 g for 30 minutes at room temperature with the brake disengaged, washed twice at 300 g for 10 minutes, cryopreserved in 10% dimethyl sulfoxide in fetal bovine serum and held at -80 C.

4. General culture conditions

Isolated CD4+ cells were cultured in RPMI GlutaMAX supplemented with 10% fetal bovine serum and 1x penicillin-streptomycin. Cells were seeded in 96-well round-bottom plates at 0.2 to 1 x 10^6 cells per well, treated with the agents stated for each study, and incubated for up to 72 hours at 37 C in 5% CO_2. All cell culture experiments included low-level IL-2 at 200 U/ml in the media. Culture conditions specific to an individual study are given in that study's section and take precedence over this paragraph.

5. Flow cytometry: staining panel and gating

Instrument	BD FACSCalibur, four-colour configuration
Preparation kit	Human Treg cell Flow Kit, used according to the instructions of the manufacturer
Surface stain	Phycoerythrin-conjugated anti-human CD25, clone BC96, detected in FL2
Intracellular stain	Alexa Fluor 488-conjugated anti-human FoxP3, clone 259D, detected in FL1, applied after fixation and permeabilisation
Suppression assay stains	Carboxyfluorescein succinimidyl ester in FL1 and allophycocyanin-conjugated anti-human CD8, clone SK1, in FL4
Conjugation	Where antibodies or fragments were conjugated in house, the Lightning-Link APC conjugation kit was used according to the instructions of the manufacturer
Resuspension	1x Hanks' balanced salt solution
Gating, population reported as Treg	Events within the CD4+ preparation that are CD25 high and FoxP3 positive, assessed on an FL2 against FL1 plot
Gating, population reported as Teff	Events within the CD4+ preparation that are CD25 high and FoxP3 negative, assessed on the same plot
Quadrant placement	Set on unstained and single-stained aliquots of the same preparation, run at the start of each acquisition session
Events acquired	A minimum of 30,000 events within the lymphocyte light-scatter region per sample
Analysis software	FlowJo version 10.0.8

6. Dose-response studies by flow cytometry

Five studies are reported in this group. Each used freshly isolated CD4+ cells prepared as in section 3.1 and cultured as in section 4. In every study each donor preparation was divided across all of the conditions listed for that study, cultured for 48 hours, then stained and acquired as in section 5. The reported quantity in every study in this group is the percentage of CD4+ events falling in the CD25 high, FoxP3 positive gate. Results are summarised as the mean with the standard error of the mean across donor preparations and are tabulated in the accompanying data files.

6.1 Study TRN416-FC-002

Donor material	Nondiabetic blood donors, freshly isolated CD4+
Samples (n)	13
Conditions	TRN-416 at 0, 0.0008, 0.004, 0.02, 0.1, 0.5, 2.5, 5, 12.5 and 25 ug/ml. Media as in section 4, including IL-2 at 200 U/ml in every well.
Incubation	48 hours
Summary statistic	Mean with SEM
Units	Percent of CD4+ events

6.2 Study TRN416-FC-003

Donor material	Donors with type 1 diabetes, freshly isolated CD4+
Samples (n)	11
Conditions	TRN-416 at 0, 0.0008, 0.004, 0.02, 0.1, 0.5, 2.5, 5, 12.5 and 25 ug/ml. Media as in section 4, including IL-2 at 200 U/ml in every well.
Incubation	48 hours
Summary statistic	Mean with SEM
Units	Percent of CD4+ events

6.3 Study TRN416-FC-004

Donor material	Nondiabetic blood donors, freshly isolated CD4+
Samples (n)	3
Conditions	TRN-416 at 0, 0.0008, 0.004, 0.02, 0.1, 0.5, 2.5, 5, 12.5 and 25 ug/ml. Recombinant human TNF at 20 ng/ml was added to every well, including the well receiving no TRN-416. Media as in section 4, including IL-2 at 200 U/ml in every well.
Incubation	48 hours
Summary statistic	Mean with SEM
Units	Percent of CD4+ events

6.4 Study TRN416-FC-005

Donor material	Nondiabetic blood donors, freshly isolated CD4+
Samples (n)	7
Conditions	Recombinant human IL-2 added at 0, 2, 10, 50, 200, 1000 and 5000 U/ml. No test article was added to any well. Doses are the quantity of IL-2 added to the well.
Incubation	48 hours
Summary statistic	Mean with SEM
Units	Percent of CD4+ events

6.5 Study TRN416-FC-006

Donor material	Nondiabetic blood donors, freshly isolated CD4+
Samples (n)	8
Conditions	Recombinant human TNF added at 0, 0.02, 0.1, 0.5, 2.5, 10, 20, 40, 80 and 100 ng/ml. No test article was added to any well. Doses are the quantity of TNF added to the well.
Incubation	48 hours
Summary statistic	Mean with SEM
Units	Percent of CD4+ events

7. Study TRN416-FC-007: absolute cell enumeration

Donor material	Nondiabetic blood donors, freshly isolated CD4+
Samples (n)	11
Conditions	TRN-416 at 0, 0.0008, 0.004, 0.02, 0.1, 0.5, 2.5, 5, 12.5 and 25 ug/ml. Media as in section 4, including IL-2 at 200 U/ml in every well.
Incubation	48 hours
Enumeration	Cells were harvested, stained as in section 5, and counted on a hemocytometer with trypan blue exclusion. Four corner squares were counted per chamber and two chambers were counted per sample.
Reported quantity	Absolute number of CD25 high, FoxP3 positive cells per well
Summary statistic	Mean with SEM
Units	Cells

8. Study TRN416-FC-008: suppression of CD8+ T cell division

Responder cells	Autologous peripheral blood mononuclear cells prepared as in section 3.2, thawed on the day before assembly and rested overnight in RPMI 1640 with 1% fetal bovine serum and IL-2 at 10 U/ml. Responders were CD8+ T cells from donors with type 1 diabetes.
Responder labelling	Carboxyfluorescein succinimidyl ester at 1 micromolar
Responder input	5 x 10^4 cells per well
Added cells	CD4+ cultures expanded with TRN-416 at 2.5 ug/ml for 48 hours. The cells added in every ratio were expanded in this way.
Ratios assembled	Responder to Treg at 0:1, 1:1, 2:1 and 4:1
Stimulation	Soluble anti-CD3 monoclonal antibody, clone HIT3a, with IL-2 at 50 U/ml in 10% fetal bovine serum RPMI
Incubation	4 days
Samples (n)	7, from normal blood donors
Readout	Percentage of CD8+-gated responder cells that underwent division, read from carboxyfluorescein succinimidyl ester dilution
Suppression index	Calculated as (Treg cell proliferation without Treg – Tresp proliferation with Treg) / Tresp proliferation without Treg cell
Summary statistic	Mean with SEM
Units	Percent of CD8+-gated events

9. Study TRN416-RS-001: transcript abundance by RNA sequencing

9.1 Cell preparation and culture

CD4+ T cells were isolated from peripheral blood by immunomagnetic negative selection using the EasySep Direct Human CD4+ T Cell Isolation Kit as described in section 3.1, following the instructions of the manufacturer. The purity of the CD4+ preparation was greater than 95%.

Cells were cultured at 37 C in 24-well tissue culture plates at 3 x 10^6 cells per well in 1 ml of RPMI with GlutaMAX, fetal bovine serum at 2% (v/v) and human recombinant IL-2 at 50 U/ml, with or without TRN-416 at a concentration of 2.5 ug/ml. Cells were then harvested and washed twice with phosphate-buffered saline, pH 7.4, calcium and magnesium free. Various incubation times were studied from 6 to 24 hours. The incubation time for the samples reported here was 10 hours.

Arms	Each donor preparation was split into two wells. One well received IL-2 at 50 U/ml and no test article. The other well received IL-2 at 50 U/ml and TRN-416 at 2.5 ug/ml.
Samples (n)	12 donor preparations
Donor material	Nondiabetic blood donors

9.2 Sample handling

Step	Condition
Harvest time after addition of test article	10 h
Interval from harvest to RNA stabilisation	1 h
Wash	Two washes in calcium- and magnesium-free phosphate-buffered saline, 300 g for 5 min
RNA isolation	RNeasy Plus Mini Kit, following the instructions of the manufacturer
Quantification	NanoDrop spectrophotometer

Step	Condition
Storage	-80 C until processed for RNA sequencing
Integrity	Agilent 2100 Bioanalyzer, RNA integrity number recorded per sample

9.3 Library preparation and sequencing

RNA sequencing was performed at two sites: the Calder Genomics Center and the Westmoor Sequencing Facility. Poly-A selection was performed using NEBNext Poly(A) mRNA Magnetic Isolation Modules, following the instructions of the manufacturer. The resulting mRNA was used for library preparation using NEBNext Ultra Directional RNA Library Prep Kits for Illumina at the Calder Genomics Center, and NEBNext Ultra II Directional RNA Library Prep Kits at the Westmoor Sequencing Facility, following the instructions of the manufacturer.

Libraries were run on high-sensitivity DNA chips on the Agilent 2100 Bioanalyzer and the functional concentration of each library was determined by quantitative polymerase chain reaction using KAPA Biosystems Illumina Library Quantification Kits. Libraries to be sequenced were pooled at a concentration of 2 nM, then denatured and diluted to a final concentration of 2 pM, and loaded onto an Illumina NextSeq 500.

The data were normalized for each sequencing laboratory separately using DESeq, part of the R-Bioconductor package.

9.4 Alignment, normalization and reporting

Alignment reference	Human genome assembly HG19
Aligner	TopHat
Transcript assembly	Cufflinks
Normalization software	DESeq, part of the R-Bioconductor package
Normalization scope	Raw reads were normalized for subjects separately.
Units	Normalized reads
Induction criterion applied	An induction of at least 1200 normalized reads was considered significant compared to cells cultured with IL-2 without antibody for the same length of time.
Analysis environment	Ingenuity Pathway Analysis software, the R Statistical Programming language and Microsoft Excel
Reported transcripts	The 70 transcripts listed in Appendix A.

10. Study TRN416-BP-009: binding kinetics by surface plasmon resonance

Service provider	Contract biophysics laboratory
Instrument	Biacore 3000
Sensor chip	BioCap
Analyte	Recombinant human TNFR2 extracellular domain
Ligand preparation	TRN-416 was biotinylated at 5:1 stoichiometry with biotinyl-LC-LC-NHS in phosphate-buffered saline. Excess biotinylating reagent was removed by centrifugation chromatography.
Capture surface	Streptavidin surface at 3000 RU
Capture level	Biotinylated antibody captured to a level of 100 RU. Antibody was injected at a concentration of 1 ng/ml to reach the final capture of 100 RU.
Theoretical maximum signal	26 RU at that level of antibody capture. That signal was reached in a preliminary experiment using 500 nM TNFR2 in the running buffer.
Flow rate	60 microlitres per minute
Injections	2 minute association injections; dissociation followed for 10 minutes
Analyte series	50, 25, 12.5, 6.25, 3.125 and 1.5625 nM
Regeneration	10 mM glycine-hydrochloride, pH 1.7, for 30 seconds
Reference channel	An identical level of streptavidin with no captured ligand
Referencing	The double reference method was used for analysis
Temperature	25 C
Fit model	1:1 Langmuir, global fit
Evaluation software	BIAevaluation 4.1.1
Dissociation constant	The K_D was 1.2(2) nM.

11. Study TRN416-BP-010: receptor counter-screen by ELISA

Format	Direct-binding enzyme-linked immunosorbent assay, 96-well

Plates	Immulon 4HBX flat bottom microtiter plates
Antigens coated	Recombinant human TNFR2 full ectodomain; the human TNFR2 fragment spanning amino acids 50 to 122; recombinant human TNFR1 full ectodomain; bovine serum albumin
Coating	1 microgram of antigen per well in carbonate coating buffer, overnight at 4 C
Primary antibody	TRN-416, eight-point 1:3 serial titration from 10 ug/ml
Secondary antibody	Horseradish peroxidase-conjugated antibody against rodent immunoglobulin G, 1:5000
Substrate	3,3′,5,5′-tetramethylbenzidine, stopped with 2 N sulfuric acid
Reader	SpectraMax 190 absorbance plate reader, 450 nm
Analysis software	SoftMax Pro 6.3
Replicates	Triplicate wells in each of three independent runs
Reported quantity	Mean optical density at 450 nm

12. Study TRN416-BP-011: epitope mapping

Linear epitope mapping on the TNFR2 external membrane protein sequence was performed by enzyme-linked immunosorbent assay. Peptides were purchased from a commercial supplier, diluted in coating buffer and placed on Immulon 4HBX flat bottom microtiter plates at a concentration of 1 mg per well. Primary monoclonal antibody at 0.1 mg per well was incubated with the substrates. Secondary antibodies against rodent immunoglobulin G were used to label the primary antibody. Absorbance was measured using the SpectraMax 190 absorbance plate reader and analysed with SoftMax Pro 6.3.

Three-dimensional peptide mapping was performed using Chemically Linked Peptides on Scaffolds (CLIPS) technology. The target protein was converted into a library of up to 10,000 overlapping linear peptides, which were then bound to a solid support and shaped into a matrix of CLIPS constructs. The affinity of the antibody to the various peptide conformations was used to determine the precise discontinuous epitopes.

Linear peptide lengths	15-mer and 18-mer, tiled at an offset of one residue
Single-loop CLIPS construct length	17-mer
Double-loop CLIPS construct length	22-mer
Primary antibody concentration	1 ug/ml, incubated overnight at 4 C
Detection	Peroxidase-conjugated antibody against mouse immunoglobulin G with 2,2′-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) substrate
Reader	Charge-coupled device camera at 620 nm
Reported quantity	Arbitrary binding units, mean of duplicate arrays
Binding sites identified	The CRD1 and CRD2 discontinuous TNFR2 binding sites were QMCCSKCSPGQH, CSSDQVET, and NRICTCRP.

13. Ancillary functional studies

13.1 Study TRN416-FC-012A: F(ab′)2 fragment

Test article	F(ab′)2 fragments of TRN-416, prepared using the Pierce F(ab′)2 Preparation Kit according to the instructions of the manufacturer and verified by size-exclusion chromatography
Donor material	Peripheral human CD4+ cells
Samples (n)	5
Conditions	F(ab′)2 at 0, 0.001, 0.0073, 0.1, 0.37, 0.91, 1.8, 2.5, 10 and 18.3 ug/ml. Media as in section 4.
Incubation	48 hours
Readout	Percent of CD4+ events in the CD25 high, FoxP3 positive gate
Summary statistic	Mean with SEM

13.2 Study TRN416-FC-012B: cross-linking antibody

Cross-linking reagent	Antibody against rodent immunoglobulin G, clone ab9165, at 2.5 ug/ml
Test article	TRN-416 at 0.0008, 0.004, 0.02, 0.1, 0.5, 2.5, 12.5 and 25 ug/ml, with the cross-linking reagent present in every well
Donor material	Peripheral human CD4+ cells
Samples (n)	3
Timepoints examined	24, 48 and 72 hours
Timepoint tabulated	48 hours
Readout	Percent of CD4+ events in the CD25 high, FoxP3 positive gate

Summary statistic	Mean with SEM

13.3 Study TRN416-EL-012C: soluble TNFR2

Assay	Quantikine enzyme-linked immunosorbent assay kit for human soluble TNFR2, with modifications
Matrix	Cell culture supernatant
Culture conditions	CD4+ cells incubated with IL-2 at 200 U/ml alone, or with TNF at 20 ng/ml, or with TRN-416 at 12.5 ug/ml
Collection window	Supernatants were collected after 24 to 42 hours of incubation
Capture	Either the commercial plates or custom plates coated with 2 ug per well of TNFR2-directed antibodies
Reader	SpectraMax 190 absorbance plate reader, 450 nm, analysed with SoftMax Pro 6.3
Samples (n)	10
Units	pg/ml

13.4 Study TRN416-AR-012D: NF-kB pathway array

Platform	Applied Biosystems TaqMan Array Human NF-kB Pathway plates, used per the instructions of the manufacturer
Culture conditions	Fresh human CD4+ T cells cultured with TRN-416 at 2.5 ug/ml, alongside wells of the same donor preparation receiving IL-2 only
Incubation	6 hours
Samples (n)	7
Gene families on the plate	The panel of assays on this plate targets genes encoding the proteins of the Rel/NF-kB family signalling pathways (NFKB1, NFKB2, RELA, RELB and REL) associated with regulatory proteins, including I-kappa-B, I-kappa-B kinase, Toll-like receptors, TNF and the TNF receptors, and TRAF; and those associated with NF-kB functions in apoptosis, immune and inflammation responses, as well as chemokines and cytokines.
Reported quantity	Relative quantitation against the wells of the same donor preparation receiving IL-2 only, which are set to 1.000
Wells without amplification	Reported as undetermined where no amplification occurred within 40 cycles

13.5 Study TRN416-MB-012E: itaconate

Matrix	Culture supernatant
Culture conditions	Fresh human CD4+ T cells cultured with TRN-416 at 0, 0.1, 0.5 and 2.5 ug/ml in 96-well plates at 100,000 cells per well. Each well in each 96-well plate yielded 50 microlitres of culture sample.
Incubation	6 hours
Donors	Supernatants were harvested from numerous normal blood donors treated in the dose-specific manner (n = 6 donors).
Analysis	Samples were sent to Metabolon Inc, Research Triangle Park, North Carolina, for the analysis of itaconate levels.
Samples (n)	12
Units	Scaled intensity

14. Statistical methods

Comparisons between two conditions	Student’s t test, two tailed, unequal variance, computed in Microsoft Excel or GraphPad Prism 8
Comparisons across more than two conditions	One-way analysis of variance with Tukey’s multiple comparison test
Dose-response trend	Linear mixed-effects modelling with participant-level random effects
Expression comparisons	Paired Student’s t test, two tailed
Summary measure	Mean with the standard error of the mean, unless otherwise stated in the study section
Software versions	GraphPad Prism 8; FlowJo 10.0.8; SoftMax Pro 6.3; BIAevaluation 4.1.1; R 3.6.1 with Bioconductor 3.9

Appendix A. Transcripts reported in study TRN416-RS-001

The following 70 transcripts were reported. Transcripts are listed in alphabetical order.

ACO1	CREB	GPI	IL7R	PGAM1	SUCLA2

ACO2	CREBBP	ID2	ITGAL	PGK1	SUCLG1
ALDOA	CS	IDH2	KEAP1	PKM	SUCLG2
BDNF	CTLA4	IDH3B	MAPRE2	PPAT	TGFB1
CAD	CXCR3	IDH3G	MDH1	PRKAA1	TNF
CCR7	ENO1	IKZF2	MDH2	RUNX1	TNFRSF18
CD274	EP300	IL10	MYC	SDHA	TNFRSF25
CD3G	ETS1	IL12B	OGDH	SDHC	TNFRSF9
CD40	FH	IL1B	PDCD1	SLC1A5	TOLLIP
CDKN1C	FOXP3	IL2	PDCD1LG2	SLC38A1	TPI1
CPT1A	GFPT1	IL2RA	PFKFB3	SLC3A2	
CPT1B	GLS2	IL4	PFKL	SLC7A5	

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